

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: Engineering Geology

Branch : Civil Engineering

Semester :

Subject Code & Name: BTCVC306_Y19 & Engineering Geology

Max Marks: 60

Date: 26-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Granite is a _____ rock.	CO1	1
	a. Plutonic, acidic b. Volcanic, acidic c. Plutonic, basic d. Volcanic, basic		
2	Rocks which are made up of one mineral are called as	CO2	1
	a. Monominerallic b. polymineralic c. both d. none of these		
3	Oceanic crust is mainly composed of -----	CO1	1
	a. Silica & aluminium b. Silica & Magnesium c. Silica & Ferrous d. Nickel & Ferrous		
4	The study of rock deformation and geometry is known as	CO2	1
	a. Structural geology b. Geophysics c. Engineering Geology d. Petrology		
5	Study of rocks as a hand specimen to identify their tentative mineral composition, chemical composition and texture is called -----	CO5	1
	a. Petrography b. Petrogenesis c. Petroscopy d. None of the above.		

6	Intermediate igneous rocks consist of ----- silica.				CO5	1
	a. Over 62%	b. About 52% to 62%	c. 45% to 54%	d. none of these.		
7	Methods used for sub-surface exploration-----				CO6	1
	a. Excavation	b. Drilling	c. Geophysical method	d. All of these		
8	In topographic maps the contour interval is dependent on-----				CO3	1
	a. Geomorphology	b. Structural defects	c. Complex geology	d. None of these		
9	The highest amount of bedload is commonly occurs in _____ stage of river.				CO3	1
	a. Youth	b. Old	c. Mature	d. Extreme Youth		
10	Drilling and blasting tunneling method is commonly preferred in ---- -- ground condition.				CO6	1
	a. Soft	b. Jointed & Stratified	c. Hard & compact	d. none of these		
11	----- dams are suitable in case of narrow river valley.				CO5	1
	a. Arch	b. Rockfill	c. Buttress	d. Gravity		
12	Which state in India has the largest dams?				CO5	1
	a. Karnataka	b. Orissa	c. Maharashtra	d. Madhya Pradesh		
Q. 2	Solve the following.					12
A)	Write a note on Interior of the earth with neat labelled diagram.				CO1	6
B)	Discuss the applications of geology in civil engineering?				CO1	6
Q.3	Solve the following.					12
A)	Describe the agents of metamorphism and their role in metamorphism.				CO2	6
B)	Define igneous rock and its classification on any one basis.				CO2	6

Q. 4	Solve Any Two of the following.		12
A)	Describe Symmetrical and Asymmetrical folds.	CO3	6
B)	Define Joints. Describe tensional joints.	CO3	6
C)	Give the six main requirements of good building stone?	CO4	6
Q.5	Solve Any Two of the following.		12
A)	Explain pros and cons of using percussion type of drilling method.	CO5	6
B)	Explain the possible adverse effects of poor geological investigation?	CO5	6
C)	Describe the limitations of drilling in the subsurface investigation.	CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the various favourable & unfavourable geological conditions for reservoir site.	CO6	6
B)	Explain in brief about rock quality designation.	CO6	6
C)	Describe exploratory drilling and core preservation.	CO6	6
	*** End ***		